

AI Exam Assistant: Intelligent Preparation & Feedback Platform

Week 09 Progress, Architecture & Full System Integration Report

COURSE TITLE	CSE4204 Mobile Computing Lab	SUBMISSION TARGET	Week 09 Progress Milestone
INSTITUTION	Northern University of Business & Technology, Khulna	COURSE INSTRUCTOR	Md. Riaz Mahmud, Assistant Professor
TEAM DESIGNATION	Team 01 (Section 8B) • Likhon Mondal (ID: 11220320865) — <i>Team Leader & Project Lead</i> • S M Tanjilur Rahman (ID: 11220320868) — <i>Backend Lead</i> • Sagor Mondal (ID: 11220320881) — <i>Frontend Lead</i> • Burhan Uddin (ID: 11220320882) — <i>AI Integration Lead</i>		
PROJECT DETAILS	Repository: https://github.com/likhon-mondal/ai-exam-assistant Live Web Portal: https://ai-exam-assistant.vercel.app		

1. Executive Summary

As of Week 09, Team 01 has completed the full-stack integration for the **AI Exam Assistant** platform. We have moved from standalone modular prototypes to a unified, production-ready web application. All core features—including JWT-secured authentication, dynamic quiz synthesis powered by Google Gemini 1.5 Flash, real-time context-aware chat assistance, PostgreSQL database persistence, and adaptive study schedule generation—are now end-to-end connected and fully functional with real data streams.

2. Feature Completion Checklist

The table below details our current feature implementation status in direct comparison with our approved Software Requirements Specification (SRS) document and project timeline.

Feature Module	Status	Technical Implementation Details	Completion Baseline
User Auth & Route Protection	COMPLETED	JWT authorization flow, bcrypt password hashing, automatic token refresh via Axios interceptors, protected client routes.	Week 07 (Completed)
AI Quiz Generation (Text & PDF)	COMPLETED	Express API endpoints connected to Gemini 1.5 Flash SDK, handling PDF uploads with Multer, Zod schema structured JSON validation.	Week 08 (Completed)
Interactive AI Study Chatbot	COMPLETED	Floating streaming chat interface in React, conversation session tracking, prompt context injection based on past quiz errors.	Week 09 (Completed)

Feature Module	Status	Technical Implementation Details	Completion Baseline
AI Personalised Study Plan	COMPLETED	Engineered prompt pipelines analyzing weak topics from PostgreSQL attempt histories to dynamically create weekly study tasks.	Week 09 (Completed)
Student Dashboard & Analytics	COMPLETED	Recharts data visualizer showing score progression, topic accuracy breakdown, attempt counts, and historical test logs.	Week 09 (Completed)
Automated Grading & Feedback	COMPLETED	Client-side quiz evaluator, server-side performance logging, step-by-step AI answer explanations stored in database.	Week 08 (Completed)
Error Log Export (PDF/CSV)	IN PROGRESS	Exporting weak topic summaries and mistake histories as downloadable reports. Core data pipeline is complete; PDF renderer in progress.	Week 10 (Target)
Peer Leaderboards & Benchmarks	NOT STARTED	Section-wide leaderboard comparisons. Deferred to prioritize individual self-assessment and core preparation workflows.	Week 11 (Target)

3. Architecture & Component Integration Status

3.1 Frontend-Backend Integration

All hardcoded static data and dummy user states in the React application have been completely replaced. Data transfer with the Express backend is managed via a centralized Axios HTTP layer featuring:

- **Bearer Token Interceptors:** Automatically attaches JWT strings to outbound HTTP headers and handles `401 Unauthorized` statuses via background session renewal.
- **Global Auth Context:** React Context API manages user authentication states, session persistence, and global user profile information.
- **Interactive Visual Feedback:** Submissions include animated loading indicators, button locking during pending async operations, and clean toast notifications.

3.2 Database Integration (PostgreSQL 15)

The backend interfaces directly with PostgreSQL 15 hosted on Render. Dynamic structured content is saved cleanly using native `JSONB` fields:

- `users`: Secure credential store, role declarations, and account metadata.
- `quizzes` & `questions`: Persists generated quiz structures, question choices, correct keys, and AI explanations in single relational payloads.
- `quiz_attempts`: Records user answers, calculated percentage scores, completion times, and identified weak topic tags.
- `chat_logs`: Stores chat session threads to maintain continuous multi-turn dialogue memory.

3.3 AI Model Integration (Google Gemini 1.5 Flash)

AI requests are executed securely through backend controller pipelines to safeguard keys and ensure standardized data formats:

- **Secure Key Storage:** Gemini API credentials are strictly isolated to server-side `.env` configurations.

- **Zod Schema Enforcement:** Model outputs are parsed using Zod schemas to guarantee strictly formatted JSON outputs free from unnecessary markdown formatting.
- **Rate Limit Mitigation:** Implemented exponential backoff middleware to manage API rate limits smoothly.

4. Major End-to-End User Workflow

The core user workflow is fully integrated and functioning smoothly end-to-end:

1. Auth: Student Register/Login (JWT Issued) & 2. **Portal:** Access Main Student Dashboard & 3. **Input:** Enter Study Topic or Drag & Drop PDF Notes & 4. **Backend Processing:** Express Controller parses file, executes Gemini 1.5 Flash prompt, validates Zod JSON schema & 5. **Storage:** Save Quiz structure into PostgreSQL & 6. **Execution:** Student takes timed adaptive quiz with instant answer feedback & AI explanations & 7. **Analytics & Planning:** Attempt saved to PostgreSQL & Dashboard Recharts updated & Dynamic Weekly AI Study Plan refreshed.

5. Integration Challenges & Technical Solutions

Technical Issue	Observed System Behavior	Implemented Solution
Markdown Invalidation	Gemini occasionally returned response strings wrapped in markdown code blocks (e.g., <code>```json ... ```</code>), causing client-side JSON parsing errors.	Added a server-side regex extraction filter combined with Zod schema validation prior to database storage or API delivery.
PDF Context Overflow	Large lecture decks exceeded default request timeouts when asking for detailed multi-question generation.	Leveraged Gemini 1.5 Flash's native file context handling combined with server-side text extraction and summarization prior to prompt execution.
JWT Session Expiration UX	Token expiration caused silent failure during quiz submission, losing student attempt data.	Implemented Axios response interceptors to catch <code>401</code> errors and retry requests after executing silent background token refreshes.
Responsive Layout Breaking	Complex quiz screens and analytics tables overflowed viewports on smaller mobile screens.	Refactored layouts using Tailwind CSS breakpoint utilities (<code>sm:</code> , <code>md:</code> , <code>lg:</code>) and wrapped data tables in horizontal scroll containers.

6. Changes & Scope Modifications

All major capabilities declared in our initial project proposal remain intact. Key technical refactorings adopted for enhanced reliability include:

- **Shift to Gemini 1.5 Flash:** Selected Gemini 1.5 Flash over earlier models to take advantage of faster generation speeds and native PDF context parsing across its large context window.
- **PostgreSQL JSONB Optimization:** Stored dynamic question choices and explanations within `JSONB` columns in PostgreSQL, significantly reducing relational query complexity.

7. GitHub Repository & Commit Best Practices

The development team practices continuous integration using feature branches. Recent commits illustrate our distributed workload:

```
# Recent Git Commit Log (Week 09 Integration Sprint) commit a8f12c4 - feat(backend): connected AI quiz generation endpoint to PostgreSQL persistence (S M Tanjilur Rahman) commit b3e4912 - feat(frontend): built streaming chatbot drawer with loading states & history (Sagor Mondal) commit c71a098 - refactor(ai): implemented Zod schema validation & backoff handler for Gemini (Burhan Uddin) commit d29f381 - docs(readme): updated architecture diagram, API routes, and Week 09 setup guide (Likhon Mondal)
```

Project Directory Structure

```
ai-exam-assistant/
├── frontend/      # React 18, Vite, Tailwind CSS, Recharts
├── backend/       # Express.js REST API, Controllers, Middleware
├── database/      # PostgreSQL schemas, SQL migrations
├── ai/            # Gemini prompt templates & Zod schemas
├── documentation/ # SRS, Design docs, Progress reports
├── screenshots/   # Interface and workflow captures
└── README.md      # Project documentation & API setup guide
```

8. System Interface & Functional Screenshots

The cards below present the core functional modules deployed and accessible in our live environment:

Module 1: Authentication & Access Control

FRONTEND / BACKEND

User registration and login interfaces featuring client-side form validation, error messaging, dark-glass UI styling, and JWT token storage.

Module 2: Main Student Performance Dashboard

FULL-STACK / DATABASE

Analytics page displaying real-time test history from PostgreSQL, Recharts score trends, topic mastery breakdowns, and single-click quiz initialization cards.

Module 3: AI Quiz Generator (Text & PDF Upload)

BACKEND / GEMINI AI

Drag-and-drop PDF uploader and topic text input interface. The backend parses documents, triggers Gemini 1.5 Flash via structured prompts, and validates JSON output schemas.

Module 4: Interactive Quiz Runner & Instant AI Explanations

FRONTEND / LOGIC

Clean single-question evaluation workspace complete with option selection, instant score calculation, question flagging, and step-by-step AI answer rationale display.

Module 5: Floating AI Study Assistant & Personalised Study Plan

AI / DATABASE

Contextual chat drawer providing instant academic guidance alongside an automatically synthesized weekly study schedule based on user weak points in the database.

Module 6: Mobile Responsiveness & Layout Adaptability

UI / UX

Fully responsive interface demonstrating mobile navigation drawers, touch-friendly option selectors, and auto-scaling chart containers across various screen viewports.

9. Individual Team Contributions

Team Member	Role	Week 09 Integration Task Contributions
Likhon Mondal (11220320865)	Team Leader	Overall system integration leadership, end-to-end user workflow validation, Week 09 progress documentation, Vercel/Render deployment configurations, and README refactoring.
S M Tanjilur Rahman (11220320868)	Backend Lead	PostgreSQL database schema migrations for quiz attempt logging and chat memory, JWT auth middleware integration, rate-limiting rules, and CORS configuration.
Sagor Mondal (11220320881)	Frontend Lead	Recharts dashboard widget integration, floating streaming chatbot UI construction, Axios request interceptor setup, and Tailwind CSS mobile layout optimizations.
Burhan Uddin (11220320882)	AI Management Lead	Gemini 1.5 Flash prompt engineering, Zod JSON schema output enforcement, automated PDF text context extraction, and API retry backoff logic.